INCH-POUND

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SUPERSEDING
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FEDERAL SPECIFICATION

ALUMINUM FOIL

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers uncoated aluminum foil.

1.2 Classification. The aluminum foil shall be supplied in the following types, classes, and grades, as specified (see 6.2):

1.2.1 Types. Aluminum foil is manufactured to the following types:

1.2.1.1 Type I. Type I aluminum foil is manufactured in rolls.

1.2.1.2 Type II. Type II aluminum foil is manufactured in interfolded flat sheets.

1.2.1.3 Type III. Type III aluminum foil is manufactured in single-ply flat sheets.

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army DEVCOM, Army Research Laboratory, Army Research Directorate, Specifications and Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A FSC 8135

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1.2.2 Classes (applicable to type II only). Type II interfolded flat sheets classes are as follows:

1.2.2.1 Class 1. Type II, class 1 aluminum foil flat sheets are manufactured to 12.0 x 10.75 inches.

1.2.2.2 Class 2. Type II, class 2 aluminum foil flat sheets are manufactured to 9.0 x 10.75 inches.

1.2.3 Grades.

1.2.3.1 Grade A. Grade A is purposed for food handling and processing applications.

1.2.3.2 Grade B. Grade B is purposed for applications other than food handling or processing.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1916	-	DoD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2. Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-123	-	Marking for Domestic Shipment (Civil Agencies)

(Copies of this document are available online at https://quicksearch.dla.mil/).

DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS)

US Code Title 21	-	Federal Food, Drug, and Cosmetic Act (FD&C Act)

(Copies of this document are available online at https://uscode.house.gov/).

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2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

THE AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes
ANSI H35.2	-	American National Standard Dimensional Tolerances for Aluminum Mill Products

(Copies of these documents are available online at https://ansi.org/).

ASTM INTERNATIONAL

ASTM B209/B209M	-	Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate Spectrometry
ASTM B479	-	Standard Specification for Annealed Aluminum and Aluminum-Alloy Foil for Flexible Barrier, Food Contact, and Other Applications
ASTM B660	-	Standard Practices for Packaging/Packing of Aluminum and Magnesium Products
ASTM D3786	-	Standard Test Method for Bursting Strength of Textile Fabrics - Diaphragm Bursting Strength Tester Method
ASTM E8/E8M	-	Standard Test Methods for Tension Testing of Metallic Materials
ASTM E1251	-	Standard Test Method for Analysis of Aluminum and Aluminum Alloys by Atomic Emission Spectrometry
ASTM E1621	-	Standard Guide for Elemental Analysis by Wavelength Dispersive X-Ray Fluorescence Spectrometry

(Copies of these documents are available online at https://www.astm.org/).

EUROPEAN COMMITTEE FOR STANDARDIZATION

EN 14242	-	Aluminium and aluminium alloys - Chemical analysis - Inductively coupled plasma optical emission spectral analysis

(Copies of this documents are available online at https://standards.iteh.ai/).

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

ISO 29681	-	Paper, board and pulps - Determination of pH of salted water extracts

(Copies of this documents are available online at https://www.iso.org/).

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3. REQUIREMENTS

3.1 Chemical composition.

3.1.1 Grade A. Grade A foil shall be produced from material of not less than 98.6 percent aluminum content, and shall contain less than 0.01 percent each of lead, arsenic, or cadmium.

3.1.2 Grade B. Grade B foil shall be produced from material of not less than 99.00 percent aluminum content.

3.2 Temper. Foil shall be in the annealed temper. In addition, for grade A foil the flexibility and strength characteristics shall be such as to permit the folding and shaping ordinarily encountered in manual and machine packaging operations and in food processing and handling without damage to the foil.

3.3 Surface finish.

3.3.1 Grade A. Grade A foil shall have a slightly slick finish.

3.3.2 Grade B. Grade B foil shall be furnished with either a dry-annealed finish or slick-annealed finish, as specified by the procuring agency (see 6.2).

3.4 Thickness and covering area. The thickness of plain foil (not interleaved) manufactured in accordance with ANSI 35.2 shall be as shown in table I. The resulting covering area is also shown.

Table I. Foil thickness and covering area.

Foil thickness (inch)	Covering area (sq. in. per pound)
0.00025	41,000
0.0003	34,200
0.00035	29,300
0.0004	25,600
0.0005	20,500
0.00065	15,800
0.0007	14,600
0.0008	12,800
0.0010	10,250
0.0015	6,830
0.0020	5,130
0.0030	3,420
0.0040	2,560

3.5 Mullen bursting strength (grade B only). The bursting strength (see 4.3.2 and 4.4.2) of grade B foil 0.0008 inch and over in thickness shall be as specified in table II.

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Table II. Bursting strength.

Nominal thickness, inch	Mullen bursting strength, psi	
	Minimum	Maximum
0.0008	8	23
0.0010	11	31
0.0015	22	55
0.0020	40	90
0.0030	75	150
0.0040	110	220
0.0050	140	280

3.6 Size.

3.6.1 Type I.

3.6.1.1 Grade A. Type I, grade A foil shall be furnished in thicknesses from 0.0007 to 0.002 inch inclusive and in widths of 12, 15, 18, or 24 inches, as specified in the contract or order (see 6.2).

3.6.1.2 Grade B. Type I, grade B foil shall be furnished in and widths as specified in the contract or order (see 6.2).

3.6.2 Type II, grade A.

3.6.2.1 Class 1. Type II, grade A, class 1 foil shall be flat sheets 12 inches by 10-3/4 inches by 0.0007 inch thick.

3.6.2.2. Class 2. Type II, grade A, class 2 foil shall be interfolded flat sheets 9 inches by 10-3/4 inches by 0.0005 inch thick.

3.6.3 Type III. The size and thickness of all type III foil shall be as specified in the contract or order (see 6.2).

3.7 Dimensional tolerances.

3.7.1 Thickness. The permissible variation of the computed thickness or measured covering area from the nominal thickness or covering area, respectively shall be not more than ±10 percent average per roll.

3.7.2 Width and length. The permissible variation in width and length of foil shall be as shown in table III.

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Table III. Permissible variation in width and length of foil.

Type and size	Tolerance (± inch)
Type I, 12 in. wide and over	1/32 1/
Type I, less than 12 in. wide	1/64 1/
Type II, classes 1 and 2	1/16
Type III, all sizes 1/16	1/16

1/ Width only.

3.8 Interleaving (type III only). Type III foil shall be interleaved with neutral paper having a pH rating in the range of 6.0 to 8.0.

3.9 Toxicity. Grade A foil shall conform in every request to the visions of the Federal Food, Drug and Cosmetic Act (FD&C Act), Food Additive Amendment, and its regulations promulgated thereunder. Grade B foil shall be free of objectional or noticeable persistent odors and free of toxic residues.

3.10 Workmanship. Foil shall be of uniform quality and condition. Defects including cuts, tears, pinholes, roughness, oxidation, smut, or contaminants that effect serviceability shall not occur in excess of the acceptable quality level (AQL) specified herein. Additional workmanship information including guidelines, definitions, and verifications is listed in ASTM B479.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. Unless otherwise specified in the contract or order, a lot shall consist of material of one type, class, grade, size, thickness, and finish offered for delivery at one time.

4.3 Sampling.

4.3.1 For chemical analysis. At least one sample shall be taken from each group of ingots of the same alloy poured simultaneously from the same source of molten metal by the producer and analyzed to determine conformance with 3.1 except that analysis for As, Pb and Cd need be made only once in each 24 hour period, provided that if the amount of As, Pb or Cd present equals or exceeds the maximum prescribed in 3.1, all samples taken during the 24 hour periods immediately preceding and following the off analysis sample shall be analyzed for the element. Ingots not conforming to the requirements of this specification shall be rejected. Complete ingot analysis records shall be available to the procuring agency at the producer's plant.

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4.3.1.1 Finished product analysis. When compliance with 4.3.1 cannot be established, one sample shall be selected from each 4000 pounds, or fraction thereof, of foil in the lot for analysis to determine conformance with 3.1.

4.3.2 For bursting strength test (grade B only). A sample for the bursting strength test (see 4.4.2) shall be taken from each 2,000 pounds or less of foil comprising the lot.

4.3.3 For determination of pH of interleaving of type III foil. One sample of interleaving paper shall be taken from each 2,000 pounds or less of type III foil comprising the lot for the purpose of determination of pH.

4.3.4 For determination of toxic residues (grade B only). Sampling for the determination of toxic residues shall be as required by the procuring agency.

4.3.5 For visual and dimensional examination. From each lot of material, samples shall be selected in accordance with MIL-STD-1916 and surveyed in accordance with ANSI/ASQ Z1.4, at inspection level I and an acceptable quality level of 6.5 percent defective. Lot size shall be expressed in terms of sheets or rolls, as applicable. For type I material, samples shall be taken from the end of each roll so selected. The samples selected for dimensional examination may be the same as those selected for visual examination.

4.4 Test procedures.

4.4.1 Chemical analysis. Chemical analysis shall be made using inductively coupled plasma atomic emission spectroscopy (ICP-AES) also known as inductively coupled plasma optical emission spectroscopy (ICP-OES) in accordance with EN 14242. Wavelength dispersive X-ray fluorescence spectrometry (XRF) in accordance with ASTM E1621 and Atomic Emission Spectrometry (AES) in accordance with ASTM E1251 are acceptable alternate methods to determine conformance to 3.1. In case of dispute, analysis using ICP-AES/OES shall be the final basis for acceptance or rejection.

4.4.2 Bursting strength test (grade B only). The bursting strength of grade B foil 0.0008 inch and greater in thickness shall be determined using the Mullen Burst test in accordance with ASTM D3786 to determine conformance to 3.5 and table II.

4.4.3 Determination of pH of interleaving. The pH of interleaving of type III foil shall be determined in accordance with ISO 29681.

4.4.4 Determination of toxic residues (grade B only). The method of determination of toxic residues shall be as required by the procuring agency. Alternatively, if acceptable to the procuring agency, a certificate of compliance shall be furnished by the producer.

4.5 Visual and dimensional examination. Each sample selected in accordance with 4.3.5 shall be subjected to surface examination for workmanship and dimensional requirements.

4.5.1 Width. The width of type I foil and the width and length of type II and III foil shall be measured.

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4.5.2 Thickness. The thickness shall be determined by weighing a known area of not less than 8 square inches on an analytical balance or other weighing device of comparable accuracy, and calculating the thickness therefrom. When the weight is ascertained in grams, the thickness in inches shall be determined by the following formula:

$$\text{Thickness in inches} = \frac{0.0226 \times \text{weight of sample in grams}}{\text{area of sample in sq. in.}}$$

4.5.3 Surface condition. When dry-annealed finish is specified (see 3.3), water shall be applied to the surface of a specimen having an area of 10 square inches or larger. Dry-annealed foil will support an unbroken water film.

4.5.4 Preparation for shipment. Examination of the packaging, packing, and marking for shipment shall be made for conformance with the requirements of section 6.

4.6 Rejection.

4.61 Examination defects. Any sample unit having one or more defects shall be rejected. If the number of nonconforming sample units in the sample exceeds the acceptance number specified in 4.3.5 for that sample size, the entire lot shall be rejected subject to the provisions of the section on "Acceptability of Lots or Batches" within ASQ/ANSI Z1.4.

4.6.2 Test failures. A lot shall be rejected for failure to meet any of the test requirements when tested in accordance with 4.4, subject to the provisions of the section on "Acceptability of Lots or Batches" within ASQ/ANSI Z1.4.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When packaging of materiel components is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.1.1 Packing considerations. Due to the substantial passage of time since the last revision of this specification, additional packaging and marking information for shipment that may be helpful including past definitions and procedures are included in 6.7 and its sections therein. This legacy content will be deleted in the next revision or amendment to this specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use.

6.1.1 Grade A. Grade A foil 0.0007 to 0.002 inch is suitable for the wrapping of foods during cooking processes. The thickness of the foil to be used will be determined by the product to be wrapped and the purpose for which it is wrapped.

6.1.2 Grade B. Grade B foil 0.0005 inch and less is suitable for aircraft applications such as the shielding or insulation of components. It is not satisfactory for fabrication of aluminum core for sandwich construction. SAE-AMS-A-81596 should be used for that purpose.

6.1.3 Thicker foils. Foils 0.0010 inch and heavier are suitable in many applications for use as a noncorrosive barrier between surfaces which have been coated with preservative compounds and wood or dunnage or between treated surfaces and other surfaces which would cause corrosive action.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Type, grade, and class required (see 1.2).
c. Finish required (see 3.3).
d. Size and thickness required (see 3.4 and 3.6).
e. Sampling and test for determination of toxic residues for grade B foil or when a certificate of compliance is acceptable (see 4.3.4 and 4.4.4).
f. Level of preservation, packaging, and packing required (sections 5.1 and 6.7).
g. Weight of foil per roll.

6.3 Covering area. The covering area is determined from the weight of a measured area of foil by the following formula:

$$\text{Covering area in sq. in./lb.} = \frac{453.6 \times \text{area of sample in sq. in.}}{\text{weight of sample in grams}}$$

6.4 Definitions.

6.4.1 Dry-annealed foil. Dry-annealed foil is foil from which the residual oil from rolling is removed in the final annealing operation.

6.4.2 Slick-annealed foil. Slick-annealed foil is foil annealed in such manner that a thin residual film of oil remains on the surface.

6.5 Precaution. Direct contact of aluminum foil with metals other than cadmium, magnesium, aluminum, or zinc, shall be avoided in applications that may be exposed to water or high humidity, to prevent galvanic action.

6.6 Supersession data. This specification includes the requirements of MIL-A-148E dated July 7, 1967.

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6.7 Preparation for delivery. Packaging for aluminum foil should follow standard section 5.1 instructions however the following legacy packing procedures have been retained due to the significant passage of years since the last revision and the wide usage of this specification.

6.7.1 Grade A.

6.7.1.1 Type I small rolls (not exceeding 5 pounds per roll).

6.7.1.1.1 Type I packaging.

6.7.1.1.1.1 Level A. Type I foil in rolls not exceeding 5 pounds in weight should be wound about paperboard cores and individually packaged in paperboard or fiberboard boxes.

6.7.1.1.1.2 Level C. Type I foil (small rolls) should be packaged in accordance with manufacturer's commercial practice.

6.7.1.1.2 Individual packaging. Packing of individually packaged rolls should be in accordance with the requirements of ASTM B660. The procuring agency should specify the level of packing required (see 6.2). Unless otherwise specified in the contract or order, material should be packed in accordance with level C.

6.7.1.1.2.1 Level C. The products should be separated by type, class, grade, and thickness and packed in accordance with the manufacturer's standard practice into containers of a type and size commonly used for the purpose, in such a manner as to ensure acceptance by carrier for transportation at the lowest rate applicable and to afford maximum protection from normal hazards of transportation.

6.7.1.2 Type I large rolls (over 5 pounds per roll) and type II. Type I (large rolls) and type II foil should be preserved and packaged in accordance with level C. Packing should be in accordance with the requirements of ASTM B660. The procuring agency should specify the level of packing required (see 6.2). Unless otherwise specified in the contract or order, material should be packed in accordance with level C.

6.7.1.2.1 Level C. See 6.7.1.1.2.1.

6.7.2 Grade B.

6.7.2.1 Type I small rods (not exceeding 5 pounds per roll). Type I (small rods) should be packaged and packed in accordance with 5.1.1.1.1.1 and 5.1.1.1.1.2, and 5.1.1.1.2.

6.7.2.1.1 Level C. See 6.7.1.1.2.1.

6.7.2.2 Type I large rolls (over 5 pounds per roll) and type III. Type I (large rolls) and type III foil should be preserved, packaged, and packed in accordance with the requirements of ASTM B660. The procuring agency should specify the level of preservation, packaging, and packing required (see 6.2). Unless otherwise specified in the contract or order, material should be preserved, packaged, and packed in accordance with level C.

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6.7.2.2.1 Level C. See 6.7.1.1.2.1.

6.8 Marking.

6.8.1 Special marking (Grade B only). All unit packages and shipping containers containing grade B foil should be conspicuously marked as follows: "NOT FOR USE WITH FOOD".

6.8.2 Civil agencies. In addition to markings required by the contract or order and 6.8.1, shipping containers shall be marked in accordance with FED-STD-123.

6.8.3 Military agencies. In addition to markings required by the contract or order and 6.8.1, shipping containers shall be marked in accordance with MIL-STD-129.

6.9 Health and safety considerations. Aluminum foil grades other than grade A should never be used for food preparation. When grade A aluminum foil is used for food preparation and is in direct contact with foods, additional care should be taken with acidic foods such as tomatoes, citrus, or vinegar based seasonings and marinades where aluminum leaching into the food is more likely to occur. Aluminum leached into foods may affect or degrade flavor and be ingested. While overall health risk from aluminum ingestion through foods is considered low, young children and sensitized adults are more likely to be affected. It is therefore good practice to limit aluminum foil contact with acidic foods, especially at elevated temperatures, or for prolonged periods of storage to minimize leaching of aluminum ions into food.

6.10 Metric units. When metric divisions are required, units for inch, foot, foot-pounds, feet per second, and pounds per square inch may be converted to the metric equivalent by multiplying them by the following conversion factors:

English	Multiply by	Equals	Metric SI unit
inch	0.0254	=	meter (m)
foot	0.3048	=	meter (m)
pound	0.4536	=	kilogram (kg)
foot-lb	1.3558	=	joule (j)
feet/second	0.3048	=	meter per second (m/s)
pounds/sq. inch	0.00689	=	Mega Pascal (MPa)
pound/cubic inch	27.6799	=	gram (g) per cubic centimeter (cm)
Fahrenheit (°F)	(t°F - 32)/1.8 = t°C		Celsius (°C)

6.11 Subject term (key word) listing.

Cooking	Food	Shielding
EMI/RFI	Food Preparation	Wrapping

6.12 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

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Preparing activity:
ARMY - MR

(Project 8135-2025-003)

Review activities:
Army - EA, MD, SM
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